Experiment 2 — Augmentation and Regularisation on Colab

This experiment isolates a stronger but still product-safe augmentation pipeline and label smoothing. It starts from the same ResNet18 checkpoint and keeps validation deterministic.

1. Configuration

```
Mode   : Local
Device : NVIDIA RTX PRO 2000 Blackwell Generation Laptop GPU
```

2. Stage images and verify the shared split

	rows	classes
train	62312	27
validation	16984	27

3. Define product-safe augmentation

4. Train only `layer4` and the classifier with regularisation

5. Run the controlled experiment

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	1	layer4_regularized	1	1.4555	1.8474	0.6103	0.6119	0.5785

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	2	layer4_regularized	2	1.3257	1.8530	0.6146	0.6132	0.5768

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	3	layer4_regularized	3	1.2296	1.8583	0.6182	0.6185	0.5801

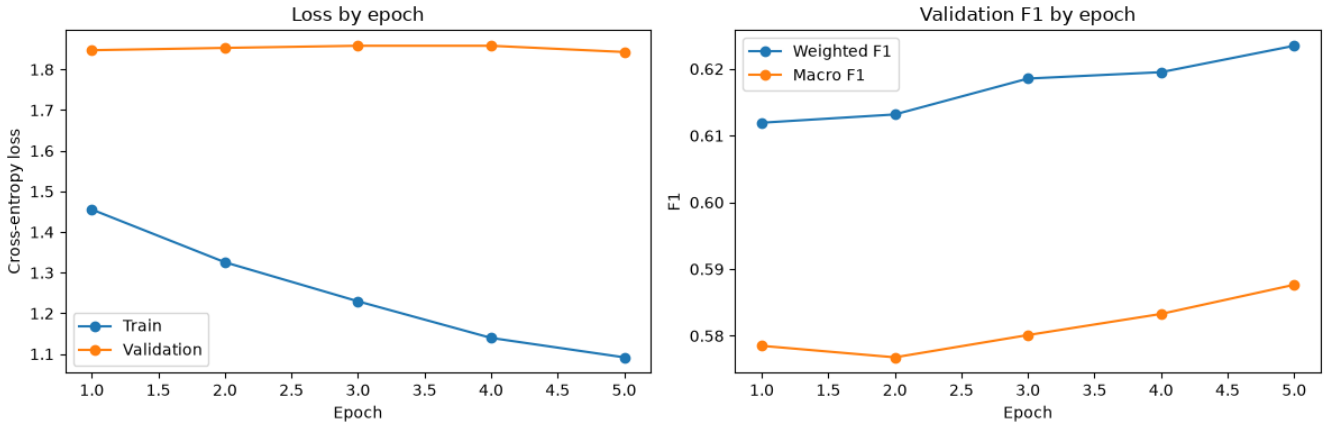
	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	4	layer4_regularized	4	1.1395	1.8583	0.6213	0.6195	0.5833

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	5	layer4_regularized	5	1.0917	1.8429	0.6245	0.6235	0.5876

6. Training curves

Use validation loss and macro F1 to assess generalisation. A falling training loss combined with a rising validation loss indicates overfitting.

	global_epoch	phase	phase_epoch	train_loss	validation_loss	accuracy	weighted_f1	macro_f1
0	1	layer4_regularized	1	1.4555	1.8474	0.6103	0.6119	0.5785
1	2	layer4_regularized	2	1.3257	1.8530	0.6146	0.6132	0.5768
2	3	layer4_regularized	3	1.2296	1.8583	0.6182	0.6185	0.5801
3	4	layer4_regularized	4	1.1395	1.8583	0.6213	0.6195	0.5833
4	5	layer4_regularized	5	1.0917	1.8429	0.6245	0.6235	0.5876



7. Evaluate and save probabilities

	validation_loss	accuracy	weighted_f1	macro_f1
resnet18_regularized	1.8429	0.6245	0.6235	0.5876

	prdtypecode	precision	recall	f1	support
8	1281	0.336	0.316	0.326	414
6	1180	0.557	0.288	0.379	153
24	2585	0.413	0.357	0.383	499
7	1280	0.405	0.387	0.396	974
17	2220	0.492	0.376	0.426	165
22	2582	0.440	0.440	0.440	518
11	1302	0.464	0.422	0.442	498
2	50	0.504	0.417	0.456	336
12	1320	0.509	0.444	0.474	648
16	2060	0.430	0.577	0.493	999
0	10	0.508	0.612	0.555	623
13	1560	0.552	0.592	0.572	1015
20	2462	0.695	0.496	0.579	284
1	40	0.637	0.546	0.588	502
19	2403	0.638	0.640	0.639	955
10	1301	0.732	0.578	0.646	161
4	1140	0.683	0.629	0.655	534
3	60	0.675	0.675	0.675	166
9	1300	0.664	0.697	0.680	1009
21	2522	0.654	0.708	0.680	998
15	1940	0.686	0.677	0.681	161
26	2905	0.777	0.661	0.714	174
18	2280	0.683	0.777	0.727	952
25	2705	0.766	0.712	0.738	552
14	1920	0.795	0.794	0.795	861
23	2583	0.851	0.808	0.829	2042
5	1160	0.884	0.914	0.899	791

Interpretation

This is a controlled comparison with the existing ResNet18. If macro F1 does not improve, reduce crop strength or remove one augmentation at a time instead of adding more operations.